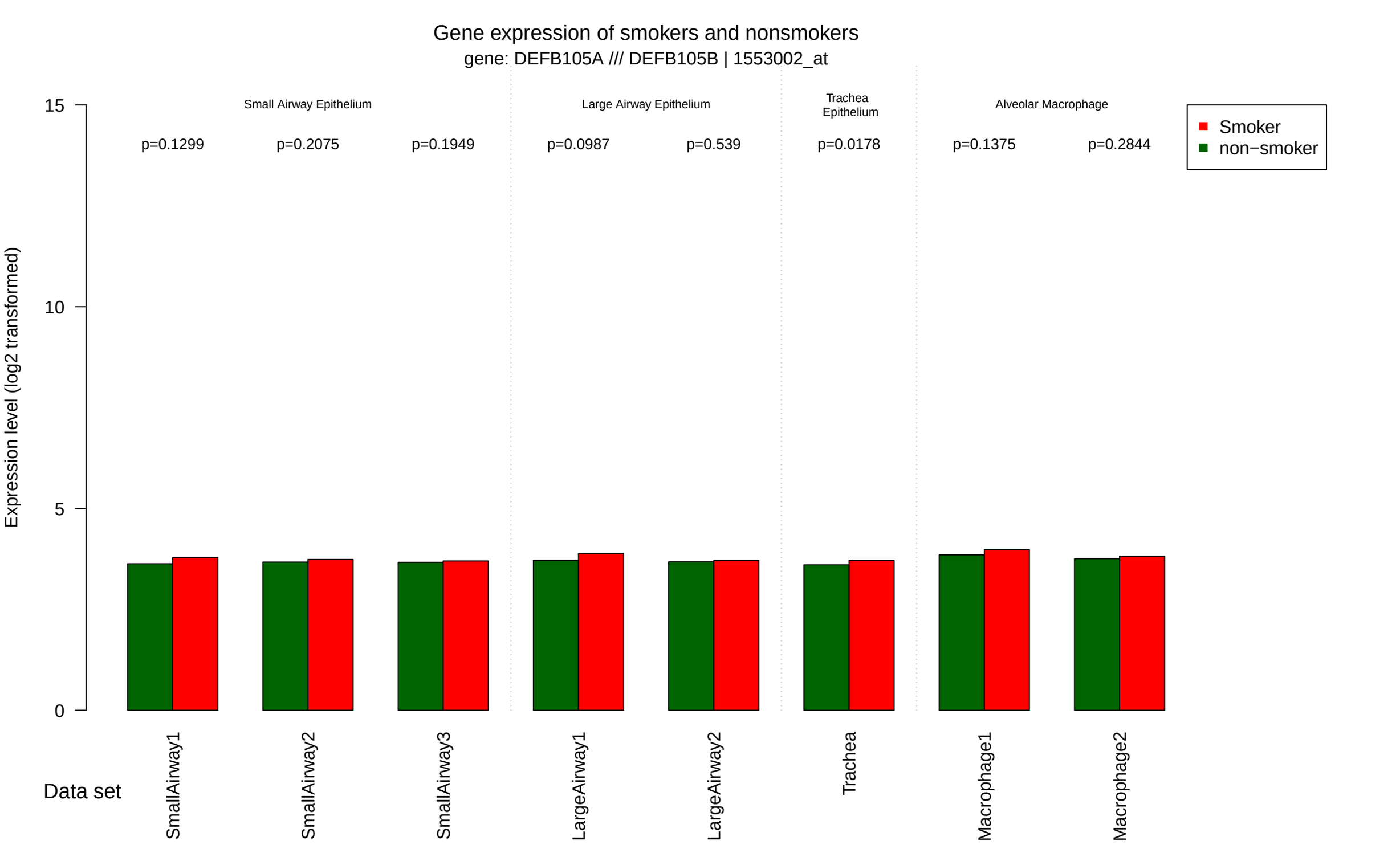
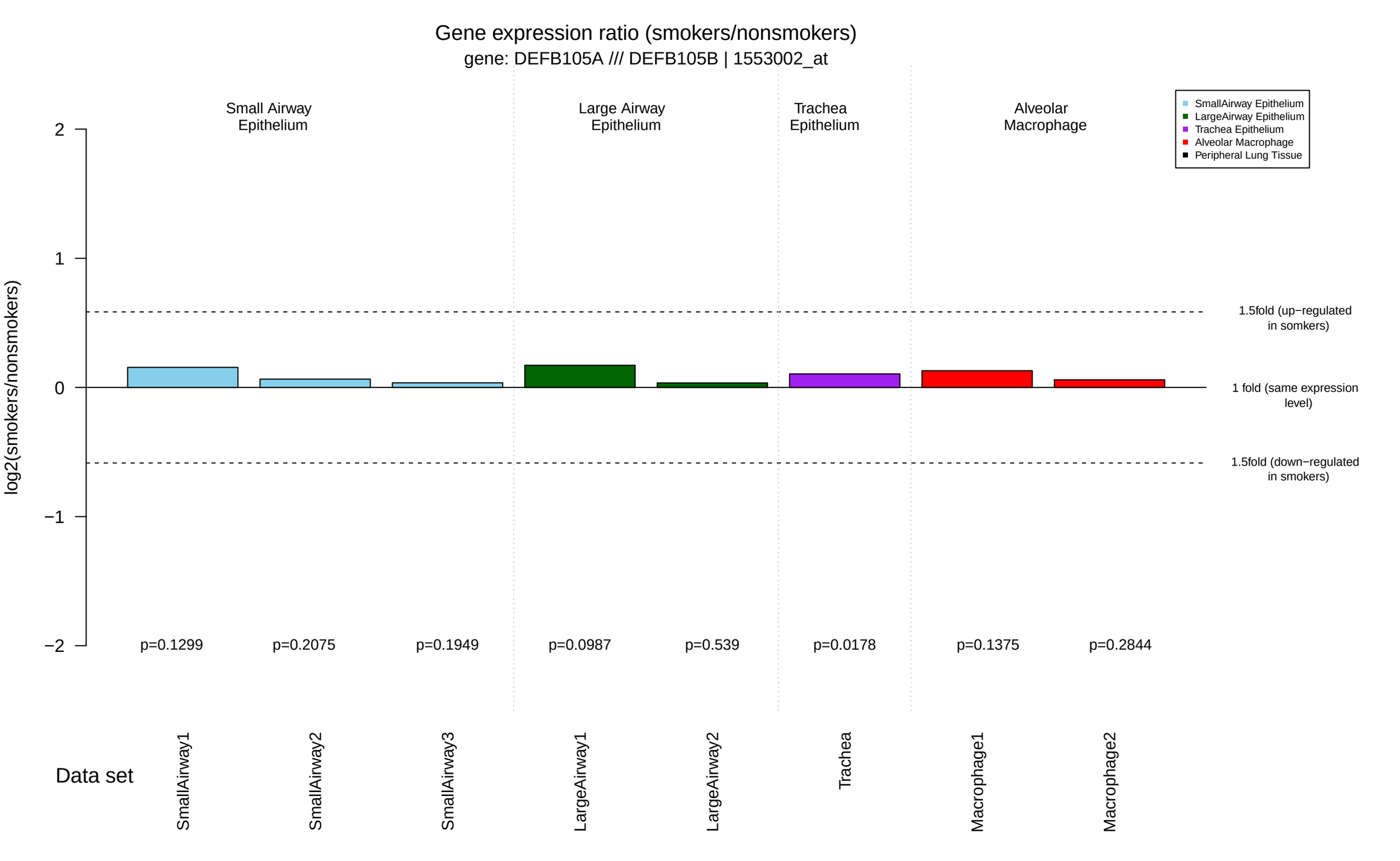
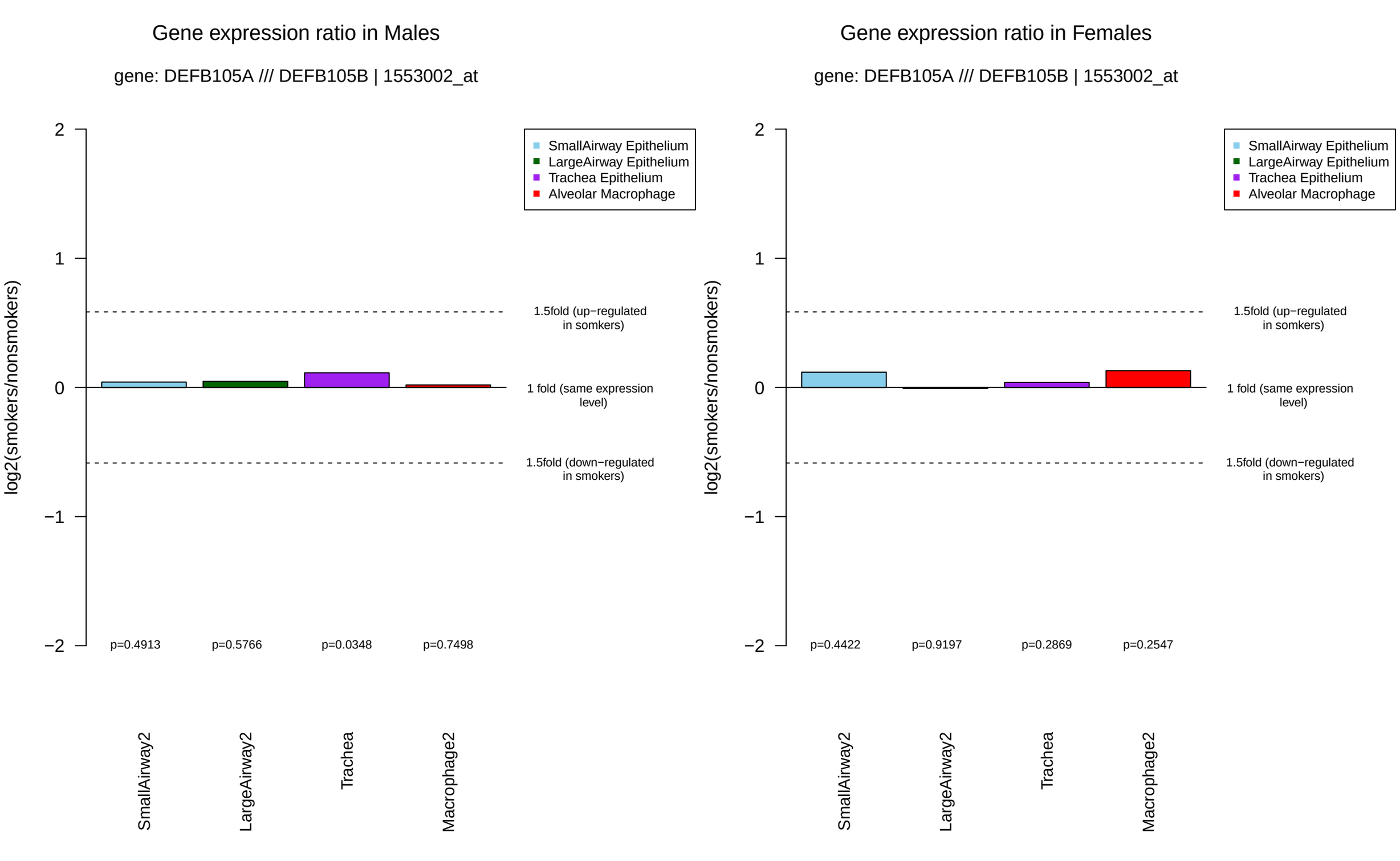
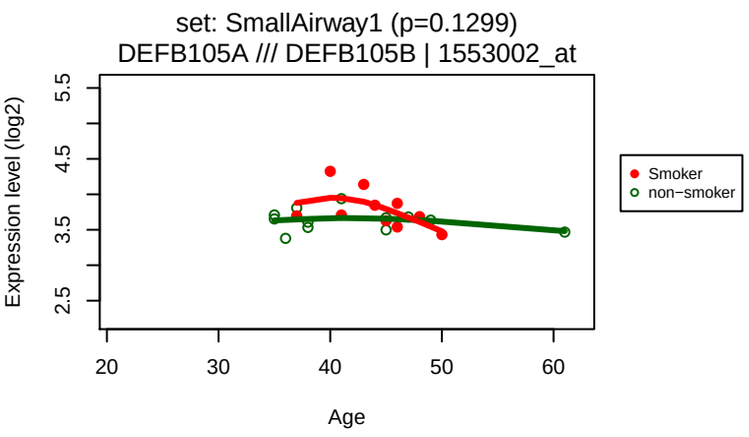
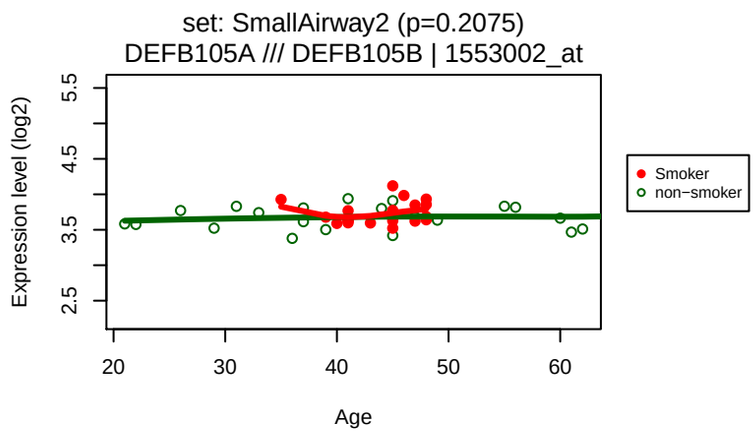
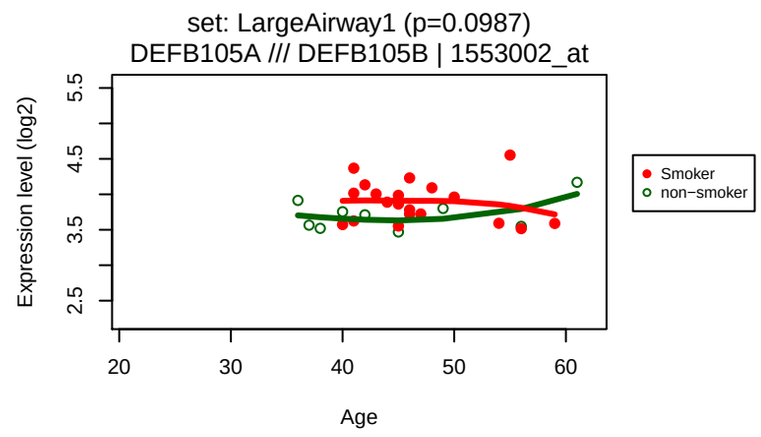

Gene expression of smokers and nonsmokers
gene: DEFB105A /// DEFB105B | 1553002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1299
p=0.2075
p=0.1949
p=0.0987
p=0.539
p=0.0178
p=0.1375
p=0.2844
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

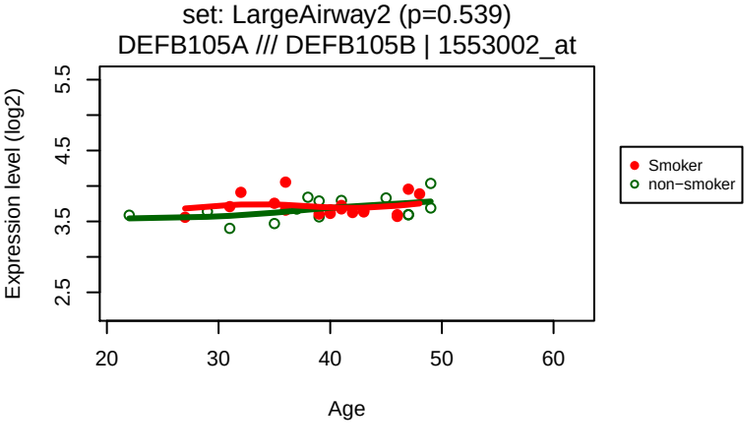

Gene expression ratio (smokers/nonsmokers)
gene: DEFB105A /// DEFB105B | 1553002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1299
p=0.2075
p=0.1949
p=0.0987
p=0.539
p=0.0178
p=0.1375
p=0.2844
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

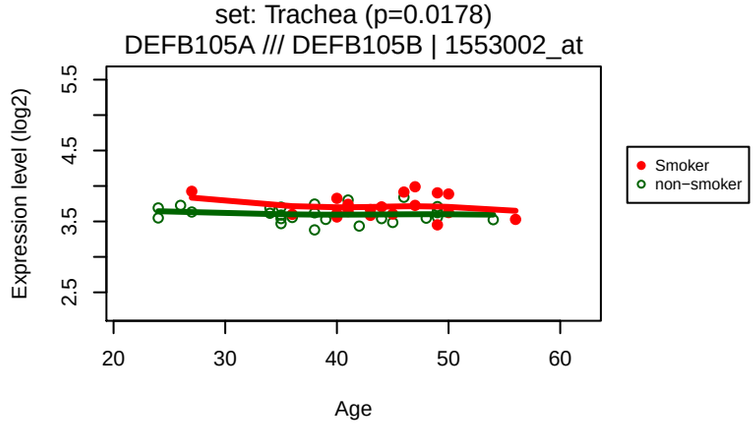

Gene expression ratio in Males
gene: DEFB105A /// DEFB105B | 1553002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4913
p=0.5766
p=0.0348
p=0.7498
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DEFB105A /// DEFB105B | 1553002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4422
p=0.9197
p=0.2869
p=0.2547
SmallAirway2
LargeAirway2
Trachea
Macrophage2

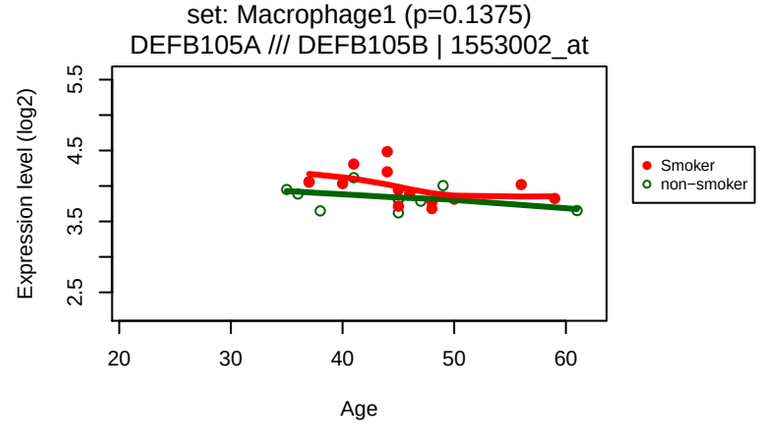

set: SmallAirway1 (p=0.1299)
DEFB105A /// DEFB105B | 1553002_at
Expression level (log2)
Age
Smoker
non−smoker

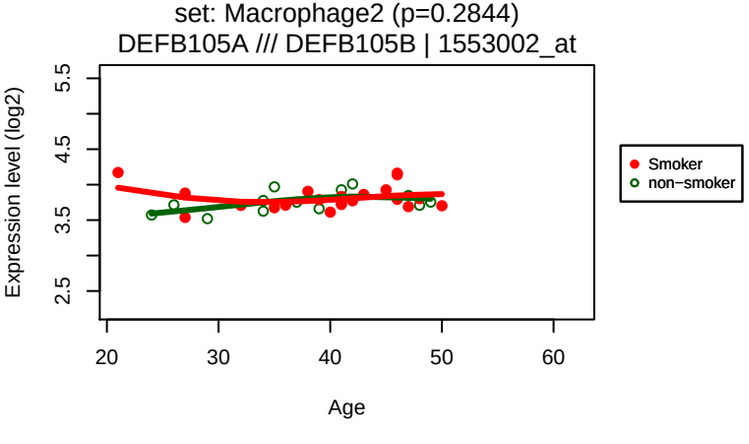

set: SmallAirway2 (p=0.2075)
DEFB105A /// DEFB105B | 1553002_at
Expression level (log2)
Age
Smoker
non−smoker

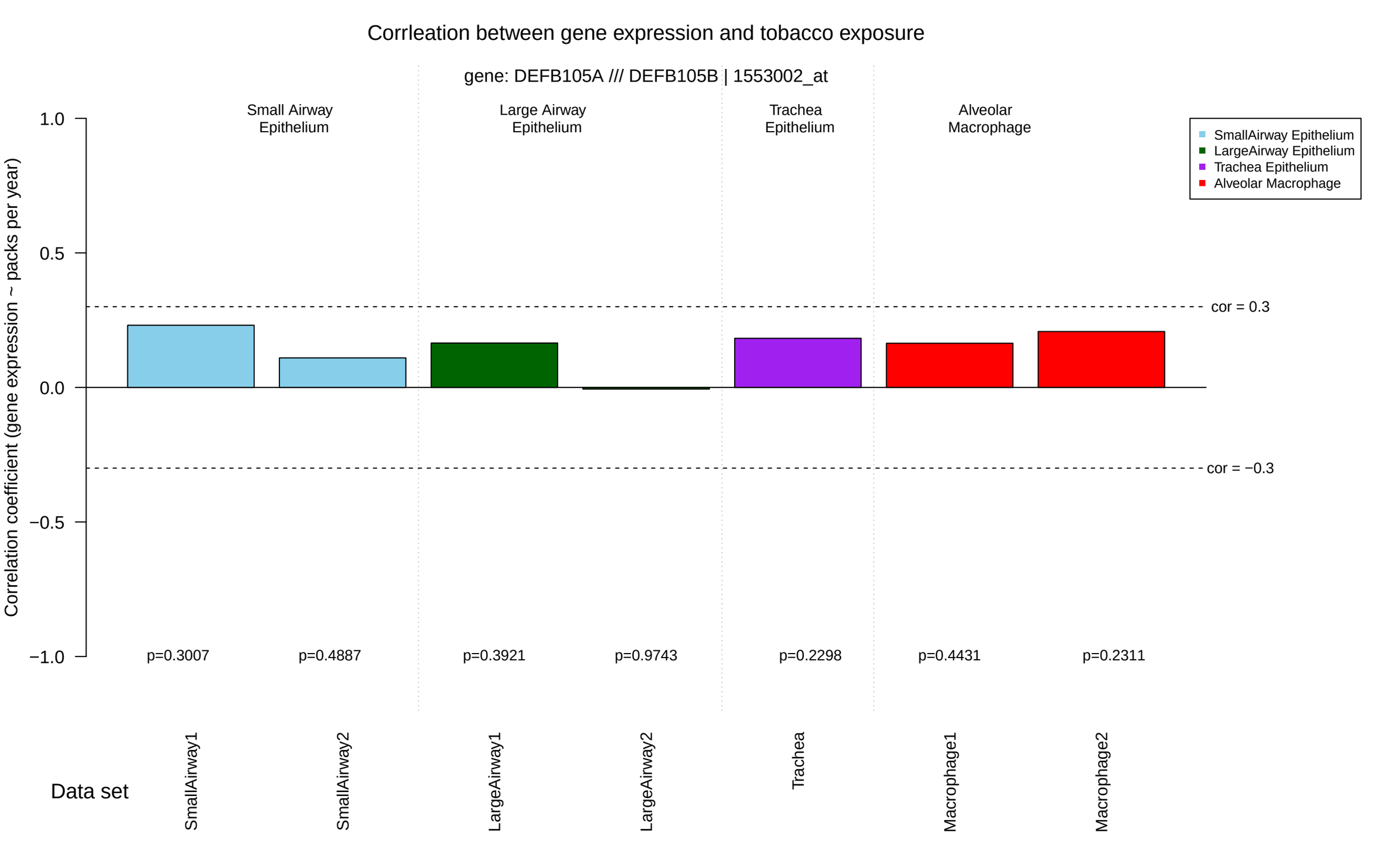

set: LargeAirway1 (p=0.0987)
DEFB105A /// DEFB105B | 1553002_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.539)
DEFB105A /// DEFB105B | 1553002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0178)
DEFB105A /// DEFB105B | 1553002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1375)
DEFB105A /// DEFB105B | 1553002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2844)
DEFB105A /// DEFB105B | 1553002_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DEFB105A /// DEFB105B | 1553002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3007
p=0.4887
p=0.3921
p=0.9743
p=0.2298
p=0.4431
p=0.2311
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2